

E-COMMERCE SALES & PROFITABILITY DASHBOARD

Power BI Dashboard Project | Multi-Platform Retail Analytics

Role: Data Analyst (Independent Project)

Tools Used: Power BI, DAX, Power Query (M), Geo-mapping (TomTom/Bing Maps)

Dataset: E-commerce order dataset — 20,000 orders across 6 platforms, 7 product categories, and multiple Indian states

Deliverable: Interactive multi-page Power BI dashboard with state/platform/category slicers, profit analysis, seasonal trends, and geo-visualization

1. Project Overview

This project analyzes order and profitability data for a multi-platform e-commerce business operating across marketplaces such as Amazon, Flipkart, Myntra, Meesho, Ajio, and Snapdeal. An interactive Power BI dashboard was built to help stakeholders track total profit, order volume, and customer demographics, while comparing performance across platforms, product categories, and geography — supporting decisions on platform investment, category focus, and regional expansion.

2. Objective

- **Consolidate** order-level e-commerce data into a single interactive view spanning state, platform, and category dimensions.
- **Measure** total profit, average profit per order, and average customer rating as core performance KPIs.
- **Compare** profit and order volume across platforms to identify the most valuable sales channels.
- **Analyze** seasonal ordering patterns across months to guide inventory and marketing timing.
- **Examine** category-level profit and order contribution to prioritize high-margin product lines.
- **Visualize** profit and order distribution geographically across Indian states.

3. Dataset & Scope

Metric	Value
Total Profit	144.74M
Total Orders	20,000 (20.00K)
Average Profit per Order	2.19K
Average Customer Rating	3.60 / 5
Platforms Covered	Amazon, Flipkart, Myntra, Meesho, Ajio, Snapdeal (6)
Product Categories	Electronics, Fashion, Home & Kitchen, Beauty & Personal Care, Sports & Fitness, Books, Grocery (7)

Metric	Value
Gender Split	Male 50.27% (10.05K orders), Female 49.73% (9.95K orders)
Geographic Scope	Multiple Indian states, mapped at state level

4. Dashboard Design

The dashboard was structured into three functional zones for guided, drill-down analysis:

- **Filter Panel (left rail):** global slicers for State, Platform, and Category, driving every visual on the page.
- **KPI & Trend Band:** four headline cards (Total Profit, Average Rating, Average Profit, Total Order), a monthly order-volume line chart, and a gender-split donut chart.
- **Analytical Core:** a horizontal bar chart ranking product categories by profit, a dual-axis platform comparison (order count vs. total profit), a vertical bar chart of order volume by category, and a geo-map layering total profit, total order, and average profit by state.

5. Key Insights

5.1 Category Profitability

- **Electronics is the top-profit category** at 143K, narrowly ahead of Fashion (130K) — together these two categories account for roughly 45% of total product-wise profit shown.
- **Grocery generates the least profit (48K)**, consistent with its typically thin retail margins, followed by Books (50K) and Sports & Fitness (57K).

5.2 Category Order Volume vs. Profit

- **Electronics leads both order volume (4.8K) and profit (143K)**, making it the strongest overall category, while Fashion is a close second on both metrics (4.4K orders, 130K profit) — confirming these two categories as the business's core revenue drivers.
- **Grocery has the lowest order volume (1.6K) and lowest profit**, suggesting it may be a lower priority category, or one sold at high volume elsewhere but underrepresented in this channel mix.

5.3 Platform Performance

- **Amazon and Flipkart show the highest combined order count and profit**, based on the dual-axis platform chart, with a declining trend across the remaining platforms (Myntra, Meesho, Ajio, Snapdeal) — indicating a concentration of business value in the top two marketplaces.
- **The order count and profit lines move closely together across platforms**, suggesting relatively consistent profit-per-order efficiency rather than one platform being disproportionately more profitable per sale.

5.4 Seasonal Ordering Trend

- **October is a major demand spike**, reaching 4.3K orders — nearly double the typical monthly baseline of ~1.1K–1.2K seen from March through September, most likely tied to festive-season and sale-event shopping (a pattern common in Indian e-commerce).

- **January also shows an elevated peak (2.7K)**, with a similar spike in June (2.7K) — both notably higher than the flat mid-year months, warranting a closer look at what campaigns or events drove those months.

5.5 Customer Demographics

- **The gender split is nearly even** (50.27% Male, 49.73% Female), indicating a balanced customer base with no single gender segment dominating purchasing behavior.

5.6 Geographic Distribution

- **Order and profit activity is concentrated in North and Central India**, based on the density of markers on the state-level map, consistent with major population and logistics hubs.

6. Business Recommendations

- Double down on Electronics and Fashion — they are the highest-profit, highest-volume categories and merit continued inventory and marketing investment.
- Prepare inventory and logistics capacity ahead of October, given the sharp seasonal demand spike, to avoid stockouts during peak sales periods.
- Reassess the Grocery category's role in the platform mix — its low margin and volume suggest either de-prioritization or a need for a different pricing/bundling strategy.
- Investigate what is driving Amazon and Flipkart's outsized share of orders and profit, and consider whether similar strategies (pricing, ads, fulfillment) can be extended to smaller platforms like Ajio and Snapdeal.
- Use the near-even gender split to inform balanced, broad-based marketing campaigns rather than skewing toward one demographic.

7. Skills Demonstrated

- Dashboard design & data storytelling in Power BI
- DAX measures for KPIs (Total Profit, Average Profit, Average Rating, Total Order)
- Data modeling across platform, category, state, and time dimensions
- Dual-axis and comparative visual design (platform profit vs. order count)
- Geo-spatial visualization (state-level mapping with profit and order overlays)
- Seasonal trend analysis and business insight generation
- Designed a dual-axis platform comparison and geo-visualization layer, revealing Amazon and Flipkart's dominant share of order volume and profit across Indian states.